

Comprehensive Mathematical Guide & Formula Compendium

Sequences and Series

Complete Formulas, Progression Rules, Operational Algorithms, and Critical Traps

1 Foundations of Sequences and Series

1.1 Definitions and General Term Analysis

- **Sequence:** A function $a : \mathbb{N} \rightarrow \mathbb{R}$ (or $\mathbb{C}$) whose domain is the set of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$. Written as $a_1, a_2, a_3, \dots, a_n, \dots$ or abbreviated as $\{a_n\}_{n=1}^{\infty}$.
- **Finite vs. Infinite Sequence:** A sequence containing a countable finite number of terms is finite; otherwise, it is infinite.
- **General Term (a_n):** The algebraic rule generating the n -th position term of the sequence.
- **Recurrence Relation:** A formula defining a_n in terms of preceding terms $a_{n-1}, a_{n-2}, \dots$ given baseline initial terms.
 - *Fibonacci Sequence:* $a_1 = 1, a_2 = 1, a_n = a_{n-1} + a_{n-2} \quad (n > 2)$.
- **Series:** The formal indicated sum associated with a sequence:

$$S = a_1 + a_2 + a_3 + \dots + a_n + \dots = \sum_{k=1}^{\infty} a_k$$

- **Fundamental Relation Between a_n and Partial Sum S_n :**

$$a_n = S_n - S_{n-1} \quad \text{for } n \geq 2, \quad \text{with } a_1 = S_1$$

When and How to Use the $a_n = S_n - S_{n-1}$ Relation

Whenever a problem gives an explicit expression for the sum to n terms $S_n = f(n)$ and asks for the n -th term a_n , the ratio of terms, or checks whether the sequence forms an A.P./G.P., immediately compute $S_n - S_{n-1}$.

Tricks & Traps: Fundamentals

1. **The $n = 1$ Boundary Trap:** $a_n = S_n - S_{n-1}$ holds **only for** $n \geq 2$. Evaluating S_0 often gives false results if the formula for S_n is not valid at $n = 0$. Always evaluate a_1 directly as $a_1 = S_1$.
2. **Series vs. Sum:** A *series* is the expression $a_1 + a_2 + \dots + a_n$. The *sum of a series* is the numerical value resulting from addition.

2 Arithmetic Progression (A.P.) and Arithmetic Mean (A.M.)

2.1 Essential A.P. Formulas

An A.P. is a sequence where $a_{k+1} - a_k = d$ (constant common difference) for all $k \geq 1$.

- **General Term (n -th term):**

$$a_n = a + (n - 1)d$$

- **n -th Term from the End:** For a finite A.P. with m terms and last term l :

$$a'_n = l - (n - 1)d = a + (m - n)d$$

- **Sum to n Terms (S_n):**

$$S_n = \frac{n}{2} [2a + (n - 1)d] = \frac{n}{2} (a + l) \quad \text{where } l = a_n = a + (n - 1)d$$

- **Properties of A.P.:**

1. If $a_1, a_2, \dots, a_n$ are in A.P., then $a_1 \pm k, a_2 \pm k, \dots$ are in A.P. (same d).
2. $ka_1, ka_2, \dots$ are in A.P. with common difference kd .
3. Sum of terms equidistant from the beginning and end is constant: $a_k + a_{n-k+1} = a_1 + a_n$.

2.2 Symmetric Term Selection in A.P.

Number of Terms	Assumed Terms	Common Difference
3	$a - d, a, a + d$	d
4	$a - 3d, a - d, a + d, a + 3d$	$2d$
5	$a - 2d, a - d, a, a + d, a + 2d$	d
6	$a - 5d, a - 3d, a - d, a + d, a + 3d, a + 5d$	$2d$

2.3 Arithmetic Mean (A.M.) and Insertion of n Means

- **Single A.M. between a and b :** $A = \frac{a+b}{2}$.
- **Insertion of n A.M.s ($A_1, A_2, \dots, A_n$) between a and b :** The sequence $a, A_1, A_2, \dots, A_n, b$ forms an A.P. with total terms $N = n + 2$.

$$b = a + (n + 1)d \implies d = \frac{b - a}{n + 1}$$

$$A_k = a + k \cdot d = a + k \left(\frac{b - a}{n + 1} \right) \quad (1 \leq k \leq n)$$

- **Theorem on Sum of n A.M.s:**

$$\sum_{k=1}^n A_k = A_1 + A_2 + \dots + A_n = n \cdot \left(\frac{a + b}{2} \right) = n \times (\text{Single A.M. between } a \text{ and } b)$$

3 Geometric Progression (G.P.)

3.1 General Term and Sum Formulas

A sequence $a_1, a_2, \dots$ is a G.P. if $a_k \neq 0$ and $\frac{a_{k+1}}{a_k} = r$ (constant common ratio) for all $k \geq 1$.

- **Standard Form:** $a, ar, ar^2, ar^3, \dots, ar^{n-1}, \dots$

- **General Term (n -th term):**

$$a_n = ar^{n-1}$$

- **n -th Term from the End:** For a finite G.P. with last term l :

$$a'_n = \frac{l}{r^{n-1}}$$

- **Sum to n Terms (S_n):**

$$S_n = \begin{cases} na & \text{if } r = 1 \\ \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1} & \text{if } r \neq 1 \end{cases}$$

- **Sum of an Infinite G.P. (S_∞):** For an infinite geometric series where $|r| < 1$ (i.e., $-1 < r < 1$), as $n \rightarrow \infty$, $r^n \rightarrow 0$:

$$S_\infty = \frac{a}{1-r} \quad (|r| < 1)$$

If $|r| \geq 1$, the infinite series does not converge to a finite sum.

3.2 Symmetric Term Selection in G.P.

Number of Terms	Assumed Terms	Common Ratio
3	$\frac{a}{r}, a, ar$	r
4	$\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$	r^2
5	$\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$	r

When to Use Symmetric Selection in G.P.

Use symmetric terms whenever the **product of terms** is explicitly provided in the problem. Multiplying the assumed terms eliminates r immediately, leaving a simple single-variable equation for a .

Tricks & Traps: Geometric Progression

1. **Infinite Sum Domain Violation:** Never apply $S_\infty = \frac{a}{1-r}$ unless you have verified $|r| < 1$. If $r = 2$, applying the formula gives $S_\infty = -a$, which is mathematically absurd for positive terms.
2. **The $r = 1$ Trap:** In parametric proofs or optimization problems, treating $S_n = \frac{a(1-r^n)}{1-r}$ without isolating the $r = 1$ case leads to illegal division by zero.
3. **Common Ratio Sign Flips:** If $r < 0$, the sequence alternates signs: positive, negative, positive, negative. Be cautious when taking roots during calculation (e.g., $r^2 = 9 \implies r = \pm 3$).

4 Geometric Mean (G.M.) and Insertion Rules

4.1 Single Geometric Mean

For two positive numbers a and b , the geometric mean G is:

$$G = \sqrt{ab}$$

The sequence a, G, b forms a G.P., since $\frac{G}{a} = \frac{b}{G} \implies G^2 = ab$.

4.2 Insertion of n Geometric Means

To insert n geometric means $G_1, G_2, \dots, G_n$ between two positive numbers a and b : The sequence $a, G_1, G_2, \dots, G_n, b$ forms a G.P. with total terms $N = n + 2$.

$$b = a \cdot r^{n+1} \implies r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$$

$$G_k = a \cdot r^k = a \left(\frac{b}{a}\right)^{\frac{k}{n+1}} \quad (1 \leq k \leq n)$$

4.3 Theorem on Product of n G.M.s

$$\prod_{k=1}^n G_k = G_1 \cdot G_2 \cdot \dots \cdot G_n = (ab)^{\frac{n}{2}} = (\sqrt{ab})^n = G^n$$

The product of n geometric means inserted between a and b equals the n -th power of the single G.M. between them.

5 Relationship Between A.M. and G.M.

5.1 The Fundamental Inequality Theorem

Let a and b be any two non-negative real numbers ($a, b \geq 0$). Let A be their Arithmetic Mean and G be their Geometric Mean:

$$A = \frac{a+b}{2}, \quad G = \sqrt{ab}$$

Then:

$$A \geq G \iff \frac{a+b}{2} \geq \sqrt{ab}$$

Condition for Equality: $A = G$ if and only if $a = b$.

5.2 Algebraic Proof

$$A - G = \frac{a+b}{2} - \sqrt{ab} = \frac{a+b-2\sqrt{ab}}{2} = \frac{(\sqrt{a}-\sqrt{b})^2}{2}$$

Since the square of any real number is non-negative, $\frac{(\sqrt{a}-\sqrt{b})^2}{2} \geq 0 \implies A - G \geq 0 \implies A \geq G$.

5.3 Constructing Quadratic Equations from A.M. and G.M.

If A and G are the A.M. and G.M. of two positive numbers a and b respectively, then a and b are the real roots of the quadratic equation:

$$x^2 - 2Ax + G^2 = 0$$

The roots are explicitly given by:

$$x = A \pm \sqrt{A^2 - G^2}$$

When and How to Use the A.M.–G.M. Inequality

Use $A \geq G$ whenever you need to find the **minimum value of a sum** of positive terms whose **product is constant** (e.g., $x + \frac{1}{x} \geq 2$), or to find the **maximum product** of terms whose sum is fixed.

General n -variable form: For positive numbers $x_1, x_2, \dots, x_n > 0$:

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n}$$

Tricks & Traps: A.M.–G.M. Inequality

1. **Non-Positive Variables Trap:** The inequality $A \geq G$ **fails** or becomes invalid if any term is negative. Always verify $x_i > 0$ before applying.
2. **Unattainable Equality:** When finding bounds using $A \geq G$, ensure that the condition $a = b$ can actually be satisfied within the domain of the variable.

6 Special Series, Algorithms, and Summation Techniques

6.1 Sum Formulas for First n Natural Numbers

1. **Sum of First n Natural Numbers:**

$$\sum_{k=1}^n k = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

2. **Sum of Squares of First n Natural Numbers:**

$$\sum_{k=1}^n k^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

3. **Sum of Cubes of First n Natural Numbers:**

$$\sum_{k=1}^n k^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2 = \left(\sum_{k=1}^n k \right)^2$$

6.2 Recurring Digit Series Algorithm

To evaluate sums of the form $S_n = d + dd + ddd + dddd + \dots$ to n terms (where $d \in \{1, 2, \dots, 9\}$):

Step-by-Step Algorithm:

1. Factor out d : $S_n = d(1 + 11 + 111 + \dots$ to n terms).
2. Multiply and divide by 9: $S_n = \frac{d}{9}(9 + 99 + 999 + \dots$ to n terms).
3. Rewrite terms as powers of 10: $9 = 10 - 1$, $99 = 10^2 - 1$, $999 = 10^3 - 1$.
4. Group into G.P. and constant terms:

$$S_n = \frac{d}{9} \left[(10 + 10^2 + 10^3 + \dots + 10^n) - \underbrace{(1 + 1 + \dots + 1)}_{n \text{ times}} \right]$$

5. Apply G.P. sum formula to obtain the general solution:

$$S_n = \frac{d}{9} \left[\frac{10(10^n - 1)}{10 - 1} - n \right] = \frac{d}{9} \left[\frac{10(10^n - 1)}{9} - n \right]$$

6.3 Method of Differences (Telescoping Series / V-N Method)

When the n -th term a_n of a series can be written as the difference of two consecutive values of a function $f(n)$:

$$a_n = f(n) - f(n-1)$$

Then the sum $S_n = \sum_{k=1}^n a_k$ collapses (telescopes):

$$S_n = [f(1) - f(0)] + [f(2) - f(1)] + \dots + [f(n) - f(n-1)] = f(n) - f(0)$$

6.4 Arithmetico-Geometric Progression (A.G.P.)

An A.G.P. is formed by multiplying corresponding terms of an A.P. and a G.P.:

$$S_n = a + (a + d)r + (a + 2d)r^2 + \dots + [a + (n-1)d]r^{n-1}$$

Reduction Method: Multiply S_n by r , shift terms right by one position, and subtract $S_n - rS_n$:

$$(1 - r)S_n = a + d(r + r^2 + r^3 + \dots + r^{n-1}) - [a + (n-1)d]r^n$$

For an infinite A.G.P. with $|r| < 1$:

$$S_\infty = \frac{a}{1 - r} + \frac{dr}{(1 - r)^2}$$

7 Master Operational Decision Tree

Problem Pattern	Recommended Method / Formula	Key Conditions & Trap Checks
Given S_n , find a_n or determine sequence type	$a_n = S_n - S_{n-1}$	Valid only for $n \geq 2$. Check $a_1 = S_1$ separately.
Given sum & product of 3 or 4 terms in A.P./G.P.	Use symmetric term assumption ($a-d, a, a+d$ or $\frac{a}{r}, a, ar$)	In 4-term setup, common difference is $2d$ and ratio is r^2 .
Sum of n inserted A.M.s	$\sum A_k = n \cdot \left(\frac{a+b}{2}\right)$	Direct multiplication; do not compute individual means.
Product of n inserted G.M.s	$\prod G_k = (ab)^{\frac{n}{2}}$	Valid for $a, b > 0$.
Finding max/min of sum of positive variables	A.M.-G.M. Inequality: $\frac{\sum x_i}{n} \geq \sqrt[n]{\prod x_i}$	Terms must be strictly positive; verify equality condition $x_i = x_j$.
Sum of repeating digits ($7 + 77 + 777 + \dots$)	$S_n = \frac{d}{9} \left[\frac{10(10^n - 1)}{9} - n \right]$	Convert to $9(10^k - 1)$ first.
Sum of series with polynomial n -th term	Express $a_n = An^3 + Bn^2 + Cn + D$; apply $\sum n^3, \sum n^2, \sum n$	Split summation linearly before applying natural number formulas.